TOWER DEFENCE LITE

Design Document

All values sourced directly from scripts

Engine	Unity 6 LTS — Universal Render Pipeline (URP)
Platform	Android — minimum API level 24 (Android 7.0)
Scenes	MainMenu	GameScene (2 scenes total)
Target FPS	30+ FPS on 3–4 GB RAM Android phones
APK Size	Maximum 50 MB
Peak RAM	Under 500 MB
Scripts	18 C# scripts across Core / Data / Enemy / Tower / Systems / UI

1. Game Overview

Tower Defence Lite is a 3D mobile tower defence game. Enemies spawn at a fixed start point and walk a waypoint path toward the player base. The player places towers on designated spots; towers automatically detect and attack enemies. The game ends in victory (all waves cleared) or defeat (base HP reaches 0).

1.1 Starting State — values from GameManager.cs

startingHP	20 → player base starts with 20 HP
startingCoins	150 → player starts with 150 coins + any saved from last session
startingGems	5 → player starts with 5 gems + any saved from last session
IsGameOver	false → set to true on defeat or victory

How coin/gem loading works (GameManager.cs Line 32–33)

Coins = startingCoins + SaveSystem.LoadCoins()
Gems = startingGems + SaveSystem.LoadGems()
SaveSystem uses PlayerPrefs key "SavedCoins" and "SavedGems" (default 0).
On first run: player gets exactly 150 coins and 5 gems.
On repeat runs: saved coins/gems stack on top of the starting amount.

1.2 Scenes & Scene Names

Scene names are hardcoded in the scripts and MUST match the Unity file names exactly:

- MainMenuUI.cs → SceneManager.LoadScene("GameScene")
- GameManager.cs → SceneManager.LoadScene("MainMenu")
- GameManager.cs → SceneManager.LoadScene(GetActiveScene().buildIndex) [Restart]

2. Enemy System

2.1 EnemyData.cs — Script Default Values

These are the default values written in EnemyData.cs. You override them in the Unity Inspector when creating ScriptableObject assets.

Field	Script Default	Type	Notes
enemyName	"Basic Enemy"	string	Pool tag = enemyName.Replace(" ","") → "BasicEnemy"
maxHealth	100f	float	Enemy health points
speed	3f	float	NavMeshAgent speed (units/sec). Set at Initialize().
damage	1	int	HP removed from player base when enemy reaches it
coinReward	10	int	Coins added via GameManager.AddCoins() on death
prefab	null	GameObject	Drag your enemy prefab here in Inspector

2.2 Enemy.cs — Key Hardcoded Values

poolTag (default)	"BasicEnemy" — must match ObjectPool tag exactly
remainingDistance check	< 0.5f → triggers move to next waypoint
pathIndex start	0 → enemies always begin at waypoints[0]
agent.isStopped on init	false → enemy starts moving immediately on Initialize()

2.3 Pool Tag Rule — from WaveManager.cs

enemyData_poolTag() extension method

Code: d.enemyName.Replace(" ", "")
Example: enemyName = "Basic Enemy" → pool tag = "BasicEnemy"
Example: enemyName = "Fast Enemy" → pool tag = "FastEnemy"
The ObjectPool Pools list tag must use this exact derived string.
Mismatch = Debug.LogWarning("Pool not found") and no enemies spawn.

2.4 EnemyHealthBar.cs

Render mode	World Space Canvas — faces camera every LateUpdate
slider.maxValue	Set to data.maxHealth in SetMaxHealth()
slider.value	Updated in UpdateHealth(float val) when TakeDamage() is called
cam reference	Auto-set to Camera.main.transform if null in LateUpdate

3. Tower System

3.1 TowerData.cs — Script Default Values

These are defaults in the script. You set final values in ScriptableObject assets in the Unity Inspector.

Field	Script Default	Type	Notes
towerName	"Tower"	string	Displayed in TowerButtonUI via towerData.towerName
damage	20f	float	Passed to enemy.TakeDamage() or Projectile.Initialize()
range	5f	float	Physics.OverlapSphere radius in Tower.FindTarget()
fireRate	1f	float	fireCooldown reset = 1f / data.fireRate
coinCost	50	int	Deducted via GameManager.SpendCoins() on placement
gemCost	0	int	If > 0: SpendGems() used instead of SpendCoins()
lockedByDefault	false	bool	If true: SaveSystem.IsTowerUnlocked() checked first
prefab	null	GameObject	Instantiated by TowerPlacementManager.TryPlaceTower()
icon	null	Sprite	Displayed by TowerButtonUI.icon Image component

3.2 Tower.cs — Key Hardcoded Values

projectilePoolTag (default)	"Projectile" — must match ObjectPool tag
fireCooldown start	0f — tower fires immediately on first valid target
fireCooldown formula	1f / data.fireRate (e.g. fireRate=1 → 1s between shots)
LayerMask	LayerMask.GetMask("Enemy") — enemy GameObjects MUST be on "Enemy" layer
Instant-hit condition	projectilePrefab == null → calls TakeDamage() directly
Projectile condition	projectilePrefab != null → Spawns from pool, calls Projectile.Initialize()
SFX key (instant hit)	"shoot" — AudioManager.Instance.PlaySFX("shoot")

3.3 TowerSpot.cs — Colors (hardcoded)

defaultColor	Color.white — spot is empty and available
hoverColor	Color.yellow — mouse is over an empty spot
occupiedColor	Color.grey — tower has been placed here
IsOccupied default	false — spot starts empty

3.4 TowerPlacementManager.cs — Placement Logic

1. Player taps TowerButtonUI button → SelectTower(data) stores the TowerData.
2. Player taps TowerSpot → OnMouseDown() → TryPlaceTower(spot).
3. Checks: selectedTowerData != null, tower is unlocked (if lockedByDefault), currency available.
4. If gemCost > 0: SpendGems(gemCost). Otherwise: SpendCoins(coinCost).
5. Instantiate(towerData.prefab, spot.position), Tower.Initialize(data), spot.PlaceTower(tower).
6. AudioManager.PlaySFX("place"). DeselectTower() clears selection.

4. Wave & Spawn System

4.1 WaveData.cs — Script Default Values

> **Bug in your script — must be fixed before use**
> WaveData currently extends MonoBehaviour but has [CreateAssetMenu] — this is wrong.
> Fix: change "public class WaveData : MonoBehaviour" to "public class WaveData : ScriptableObject"
> Without this fix you cannot create WaveData assets in the Unity Project window.

WaveEntry.count	5 — default enemies per entry (set in Inspector per wave)
WaveEntry.spawnInterval	1f — seconds between each enemy spawn within one entry
timeBetweenWaves	5f — seconds of rest after a wave ends before next begins

4.2 WaveManager.cs — Hardcoded Values

Initial delay	WaitForSeconds(2f) — 2 seconds before Wave 1 begins
CurrentWave start	0 — becomes 1 when first wave starts (w + 1)
enemiesAlive start	0 — incremented on SpawnEnemy(), decremented on OnEnemyDefeated()
Between-wave wait	waves[w].timeBetweenWaves — read from WaveData asset
Victory condition	All waves loop complete + IsGameOver == false → TriggerVictory()
Defeat check	GameManager.IsGameOver == true → yield break (stops all spawning)

4.3 ObjectPool.cs — Default Values

Pool.size default	10 — pre-creates 10 instances of each pool at scene load
Dynamic expansion	If queue.Count == 0: Instantiate() a new one (safety fallback)
Spawn()	Dequeues, sets position/rotation, SetActive(true)
ReturnToPool()	SetActive(false), Enqueues back — zero heap allocation

5. Save System

5.1 SaveSystem.cs — PlayerPrefs Keys & Defaults

Method	PlayerPrefs Key	Default Value	Called By
SaveCoins(v) / LoadCoins()	"SavedCoins"	0	GameManager.SpendCoins, AddCoins
SaveGems(v) / LoadGems()	"SavedGems"	0	GameManager.SpendGems, AddGems
SaveBestWave(v) / LoadBestWave()	"BestWave"	0	GameManager.DamageBase, TriggerVictory
UnlockTower(name)	"TowerUnlocked" + name	0 (locked)	ShopUI buy button listener
IsTowerUnlocked(name)	"TowerUnlocked" + name	0→false	TowerPlacementManager.TryPlaceTower
Save()	PlayerPrefs.Save()	—	ShopUI after tower purchase

6. UI System

6.1 UIManager.cs — All Inspector Fields

Every field below must be assigned in the Unity Inspector. Null fields cause NullReferenceException on Start().

Header Group	Field Name	Type	What to drag
HUD	coinText	TextMeshProUGUI	Coin count text in HUD
HUD	gemText	TextMeshProUGUI	Gem count text in HUD
HUD	hpText	TextMeshProUGUI	HP/lives text in HUD
HUD	waveText	TextMeshProUGUI	Wave number text in HUD
Panels	pausePanel	GameObject	PausePanel (starts inactive)
Panels	gameOverPanel	GameObject	GameOverPanel (starts inactive)
Panels	victoryPanel	GameObject	VictoryPanel (starts inactive)
Panels	shopPanel	GameObject	ShopPanel (starts inactive)
Panels	settingsPanel	GameObject	SettingsPanel (starts inactive)
Panels	messagePanel	GameObject	MessagePanel (starts inactive)
Panels	messageText	TextMeshProUGUI	Text inside MessagePanel
Result Screen	resultWaveText	TextMeshProUGUI	Wave Reached text in result panels
Result Screen	resultEnemiesText	TextMeshProUGUI	Enemies Defeated text

Header Group	Field Name	Type	What to drag
Result Screen	resultCoinsText	TextMeshProUGUI	Coins Earned text
Settings	musicToggle	Toggle	Music on/off Toggle
Settings	sfxToggle	Toggle	SFX on/off Toggle
Leaderboard	bestWaveText	TextMeshProUGUI	Shows "Best: Wave X"

6.2 UIManager.cs — Button OnClick Methods

Button	Method to wire	Effect
Pause Button	OnPauseButton()	Toggle PausePanel + Time.timeScale 0/1
Resume (in Pause)	OnResumeButton()	Hide PausePanel, timeScale = 1
Restart (in Pause)	OnRestartButton()	timeScale=1, reload scene by buildIndex
Main Menu (in Pause)	OnMainMenuButton()	timeScale=1, LoadScene("MainMenu")
Restart (GameOver)	OnRestartButton()	Same as above
Main Menu (GameOver)	OnMainMenuButton()	Same as above
Shop Button (HUD)	OnShopButton()	Toggle ShopPanel SetActive
Settings Button	OnSettingsButton()	Toggle SettingsPanel SetActive
Music Toggle	OnMusicToggle(bool)	AudioManager.ToggleMusic()
SFX Toggle	OnSFXToggle(bool)	AudioManager.ToggleSFX()

6.3 Message Toast — hardcoded duration

ShowMessage(msg)	Sets text, SetActive(true), CancelInvoke, Invoke(HideMessage, 2f)
Hide delay	2 seconds — hardcoded Invoke(nameof(HideMessage), 2f)
Auto-dismiss	No player interaction needed — disappears automatically

7. Audio System

7.1 AudioManager.cs — SFX Keys Used in Scripts

SFX Key (string)	Called From	When
"shoot"	Tower.cs Shoot() — instant-hit mode only	Tower fires without a projectile
"hit"	Projectile.cs Update() — on proximity hit	Projectile reaches enemy
"place"	TowerPlacementManager.cs TryPlaceTower()	Tower successfully placed
"click"	TowerButtonUI.cs OnClick()	Tower selection button tapped

7.2 AudioManager.cs — Saved Settings Keys

PlayerPrefs key (music)	"MusicOn" — default 1 (on). Loaded in Awake().
PlayerPrefs key (SFX)	"SFXOn" — default 1 (on). Loaded in Awake().
DontDestroyOnLoad	AudioManager persists across scene changes (MainMenu → GameScene)
musicSource	AudioSource: Loop=true, PlayOnAwake=true (background music)
sfxSource	AudioSource: PlayOneShot() — allows overlapping sounds

8. Known Bugs — Fix Before Building

These 5 issues exist in your scripts and will cause errors:

1. WaveData.cs → change ": MonoBehaviour" to ": ScriptableObject" (line 3)
2. TowerData.cs → delete "using UnityEditor.ShaderGraph.Internal;" (line 1) — breaks Android build
3. ObjectPool.cs → delete "using UnityEngine.InputSystem.Controls;" — unused, causes errors
4. EnemyHealthBar.cs → delete "using UnityEditor;" — breaks Android build
5. WaveManager.cs → delete "using Unity.VisualScripting;" — only if package not installed

9. MainMenu Scene

9.1 MainMenuUI.cs — Button Methods

OnPlayButton()	SceneManager.LoadScene("GameScene") — loads game scene
OnSettingButton()	settingPanel?.SetActive(!settingPanel.activeSelf) — toggles settings panel
OnQuitButton()	Application.Quit() — closes the app

settingPanel field	public GameObject — drag your Settings Panel here in Inspector

End of Design Document | All values verified against script source